

Jiaqi Ye

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EDUCATION

University of Wisconsin-Madison | Madison, WI, USA

Candidate for B.S. Computer Science | Jan. 2024 – Expected Dec. 2026

GPA: 3.96/4.0. Honors: Dean's List for Six Consecutive Terms; Undergraduate Scholarship for Summer Study

Beijing Forestry University | Beijing, China

Undergraduate Major: Digital Media Art | Sep. 2021 – Dec. 2023

GPA: 3.8/4.0. Honors: Academic Scholarships, College Outstanding Student Leader

INTERNSHIPS

UW Surgery | Madison, USA

AI Engineer | Jan. 2026 – Present

- Built a guideline-grounded **RAG** system with a **PostgreSQL** retrieval **API**, indexing 8 clinical reference documents into 289 chunks with citations.
- Improved retrieval accuracy by 15% via semantic chunking, embedding-based similarity search, automated deduplication, and top-12 evidence retrieval.
- Created a benchmark suite evaluating five **LLMs** on 50 patient-style FAQs, using **Llama 3.1 8B Instruct** as an automated judge for accuracy and clarity.
- Implemented a multi-agent safety pipeline for triage, intent analysis, and scope validation, with prompt-injection defenses and schema checks.

The Donovan's Venom | Los Angeles, USA

Full-Stack Developer | Apr. 2025 – Aug. 2025

- Developed full-stack features across a **Next.js/React/TypeScript** frontend and **Bun/Elysia** backend with **PostgreSQL** for a music education platform.
- Implemented authentication and session management across the platform, securing checkout and account workflows for signed-in and guest visitors.
- Shipped signed-in and guest checkout flows integrating **PayPal** and **Venmo**, reducing transaction errors by 25% across the entire payment pipeline.
- Redesigned the Cart and Contact pages for responsive UX, cutting page load time by 30% through optimized asset loading and layout redesign work.

SELECTED PROJECTS

GitLab Engineering & Security Analytics Platform | FastAPI, React, MySQL, AWS EC2, GitLab CI/CD, GPT

- Deployed a full-stack analytics platform with **FastAPI**, **React**, and **MySQL** on **AWS EC2**, exposing **RESTful APIs** to ingest data across 6 domains.
- Engineered an incremental analytics pipeline with pagination, deduplication, and rate-limit handling to track 7+ KPIs, including merge latency.
- Monitored **GitLab CI/CD** pipeline reliability and remediation trends across projects, surfacing recurring build failures and stale backlog items.
- Integrated a **GPT**-powered insights layer grounded in computed project metrics to identify development trends and workflow bottleneck signals early.

Labfy | First Prize, MIT CSAIL Agentic AI Hackathon | Python, FastAPI, Chroma, arXiv API

- Engineered a multimodal AI lab copilot integrating smart-glasses voice and image capture with transcription and vision analysis for hands-free use.
- Developed a **Python/FastAPI** agent backend exposing **RESTful** endpoints that orchestrate 19 tools across procurement, notes, and manuscript drafting.
- Implemented human-in-the-loop procurement workflows validating grant eligibility, budgets, and spending caps before routing purchases for approval.
- Assembled a grounded manuscript-generation pipeline synthesizing lab notes with **Chroma**-retrieved papers and live **arXiv** results to draft methods.

Lab Animal Procurement Agent | Full-Stack AI Agent System | Python, FAISS, AWS S3, Docker Compose

- Owned an AI-powered procurement agent for lab animal management, supporting strain selection, cage capacity checks, and vendor comparison tools.
- Sourced SOPs and papers from **S3** for a **PDF**-grounded **RAG** pipeline, using semantic chunking, embedding generation, and **FAISS**-based retrieval.
- Designed vendor, strain, and cage management **RESTful APIs** with profile-aware capacity constraints and automated RFQ and order email drafting flows.
- Containerized the full stack with **Docker Compose** and added automated regression testing for core API endpoints and multi-agent workflows end to end.

Lab Equipment Monitoring & Alerting Platform | Python, Kafka, Redis, Kubernetes (EKS), AWS Lambda, API Gateway

- Streamed lab-equipment sensor readings through **Kafka** topics, publishing temperature and usage events at sub-second latency across 30+ instruments.
- Cached device state and threshold configs in **Redis**, cutting p95 alert-evaluation latency under bursty traffic from dozens of monitored lab devices.
- Deployed the alerting service on a **Kubernetes (EKS)** cluster with rolling updates, ensuring zero downtime during on-call configuration changes.
- Exposed **AWS Lambda** functions behind **API Gateway** for on-demand equipment-status queries from the lab's mobile app, autoscaling with query volume.
- Coordinated the buildout with the lab's engineering team in **Agile/Scrum** sprints, delivering the alerting MVP across three two-week iterations.

TECHNICAL SKILLS

Programming & Databases: Python, Java, C#, C/C++, JavaScript/TypeScript, SQL, PostgreSQL, MySQL

AI/ML & Agent Systems: PyTorch, Hugging Face Transformers, RAG, LLM Agents, Tool Calling, Chroma, FAISS, OpenAI API

Web & Backend Development: Spring Boot, FastAPI, React, Next.js, Node.js, REST APIs, Prisma, Vite

Cloud & Infrastructure: AWS (EC2, S3, Lambda, API Gateway), Kubernetes (EKS), Docker, GitLab CI/CD, Kafka, Redis

Tools & Design: Git, Linux, Agile/Scrum, Unity, Blender, Figma, Adobe Creative Suite